

$$\begin{aligned}
& \mu \tilde{g}_1^4 y_d^{i1i2} \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{12} m_2 \tilde{\mu} \tilde{g}_2^4 (9 y_d^{i1i2} + 4 y_d^{pi2} \delta_{pi1}) \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \\
& g_2^4 y_d^{pi2} \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \delta_{pi1} + \frac{1}{18} \tilde{\mu} \tilde{g}_1^2 y_d^{pr} \overline{a_d^{pr}} (-3 y_d^{i1i2} + y_d^{s12} \delta_{s11} + 2 y_d^{i1s} \delta_{s12}) \\
& \times [\mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{36} \tilde{\mu} \tilde{g}_1^2 y_d^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \\
& 2 y_e^{pr} \overline{a_d^{pr}} (-3 y_d^{i1i2} + y_d^{s12} \delta_{s11} + 2 y_d^{i1s} \delta_{s12}) \text{LF}_{2,2,0}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& 2 y_e^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& 2 y_e^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& 2 y_e^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_q^p] + \frac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_d^{pr}} \\
& (7 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + g_1^2 (-3 y_d^{i1i2} + y_d^{s12} \delta_{s11} + 2 y_d^{i1s} \delta_{s12})) \text{LF}_{4,1,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& y_d^{pr} \overline{a_d^{pr}} (3 g_2^2 (6 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + g_1^2 (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12})) \\
& \times [\mathbf{m}_l^p, \mathbf{m}_e^r] + \frac{1}{18} \tilde{\mu} \tilde{g}_1^2 y_d^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& 2 y_d^{pr} \overline{a_d^{pr}} (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& y_d^{pr} \overline{a_d^{pr}} (-9 g_2^2 (7 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + 5 g_1^2 (-3 y_d^{i1i2} + y_d^{s12} \delta_{s11} + 2 y_d^{i1s} \delta_{s12})) \\
& \times [\mathbf{m}_q^p, \mathbf{m}_d^r] + \frac{1}{6} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} \\
& (6 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + g_1^2 (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12})) \text{LF}_{5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& y_d^{pr} \overline{a_d^{pr}} (- (g_2^2 + 9 g_2^2) (3 y_d^{i1i2} - y_d^{s12} \delta_{s11}) + 2 g_1^2 y_d^{i1s} \delta_{s12}) \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \\
& y_d^{pr} \overline{a_d^{pr}} (-9 g_2^2 (6 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + g_1^2 (-3 y_d^{i1i2} + y_d^{s12} \delta_{s11} + 2 y_d^{i1s} \delta_{s12})) \\
& \times [\mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{6} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} \\
& (6 y_d^{i1i2} + y_d^{s12} \delta_{s11}) + g_1^2 (3 y_d^{i1i2} - y_d^{s12} \delta_{s11} - 2 y_d^{i1s} \delta_{s12})) \text{LF}_{6,1,-2}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \\
& g_1^2 g_2^2 y_d^{i1i2} \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_1] + \frac{7}{8} m_1 \tilde{\mu} \tilde{g}_1^4 y_d^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] + \\
& g_1^2 (-3 g_2^2 (12 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{ip1} \delta_{pi2})) \\
& \times [\tilde{\mu}, \mathbf{m}_1] - \frac{1}{2} m_1 \tilde{\mu} \tilde{g}_1^4 y_d^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\
& g_1^2 (3 g_2^2 (6 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (3 y_d^{i1i2} - y_d^{pi2} \delta_{pi1} - 2 y_d^{ip1} \delta_{pi2})) \\
& \times [\tilde{\mu}, \mathbf{m}_1] + \frac{1}{6} m_2 \tilde{\mu} g_2^4 (9 y_d^{i1i2} + 2 y_d^{pi2} \delta_{pi1}) \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_2] + \\
& g_2^4 (57 y_d^{i1i2} + 8 y_d^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] + \\
& g_2^2 (-g_2^2 (36 y_d^{i1i2} + 7 y_d^{pi2} \delta_{pi1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{ip1} \delta_{pi2})) \\
& \times [\tilde{\mu}, \mathbf{m}_2] - 2 m_2 \tilde{\mu} g_2^4 y_d^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \\
& g_2^2 (3 g_2^2 (6 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (3 y_d^{i1i2} - y_d^{pi2} \delta_{pi1} - 2 y_d^{ip1} \delta_{pi2})) \\
& \times [\tilde{\mu}, \mathbf{m}_2] + \frac{1}{36} m_1 \tilde{\mu} \tilde{g}_1^2 \overline{y_d^{pi2}} y_d^{i1r} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^p] - \\
& g_1^2 \overline{y_d^{pi2}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^p] - \\
& g_1^2 \overline{y_d^{pi2}} y_d^{pi2} y_d^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^p] + \\
& g_1^2 g_2^2 y_d^{i1i2} \text{LF}_{2,2,-1,-1}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \mathbf{m}_q^{i1}] + \frac{2}{3} m_2 \tilde{\mu} \tilde{g}_1^4 y_d^{i1i2} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] - \\
& g_1^4 y_d^{i1i2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] + \frac{1}{72} m_1 \tilde{\mu} \tilde{g}_1^2 \overline{y_d^{pi2}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p, \mathbf{m}_d^{i2}] + \\
& g_1^2 \overline{y_d^{pi2}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_q^p, \tilde{\mu}] - \\
& g_1^2 \overline{y_d^{pi2}} y_d^{pi2} y_d^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p, \tilde{\mu}] + \\
& g_1^2 g_2^2 y_d^{i1i2} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_d^{i2}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \\
& \times [\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_d^r] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_d^r] + \\
& g_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \mathbf{m}_d^r] + \frac{1}{18} m_1 \tilde{\mu} \tilde{g}_1^4 y_d^{i1i2} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \\
& g_1^4 y_d^{i1i2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \frac{1}{36} m_1 \tilde{\mu} \tilde{g}_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_d^r, \mathbf{m}_q^{i1}] + \\
& g_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^r, \tilde{\mu}] - \\
& g_1^2 y_d^{pi2} \overline{y_u^{i1r}} y_u^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^r, \tilde{\mu}] - \\
& g_1^2 (3 y_d^{pr} y_d^{pi2} y_d^{i1r} + y_d^{i1i2} (14 g_1^2 + 3 g_2^2)) \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_d^{i2}] - \\
& g_1^2 \overline{y_d$$